AI Based-Tourism Safety and Itinerary Platform for Sri Lanka

Project ID: R26-IT-152

De Silva D.S.K (IT22108654) | Save Our Souls Emergency Detection	Wanniarachchi P.M.R (IT22225092) | AI Itinerary Generator	De Silva D.C.M (IT22108586) | Hotel-Flight Driver Matcher	Abinaya R (IT22090058) | Tourism Risk and Context Intelligence



Supervisor: Mr. Buddika Harshanath | **Co-Supervisor:** Prof. Samantha Rajapaksa

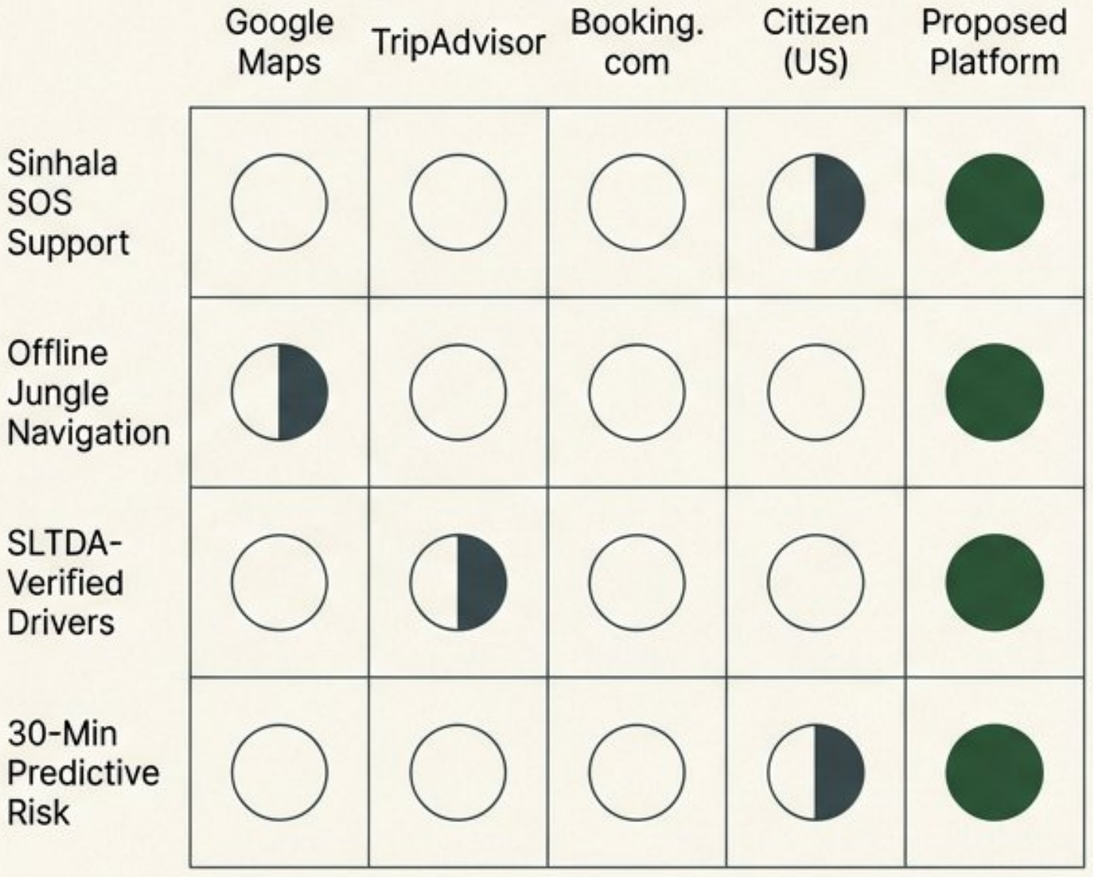
Centre of Excellence for Artificial Intelligence | SLIIT | March 2026

Five Documented Infrastructure Failures Endangering Visitors

The Stakes

1. **2,360,000** arrivals in 2025, yet a **$1.6 Billion** annual revenue gap directly tied to safety perception deficits.
2. More than **200 fatalities** annually from road accidents and lack of real-time emergency response.
3. A critical shortage of **1,600+** registered guides.
4. **Zero** site-level risk prediction across **500+** registered destinations.

App Ecosystem Matrix



Four Coordinated Pillars Forming a Unified Ecosystem

The Ear and Voice (De Silva D.S.K)

Multilingual SOS distress detection, 30-minute danger zone prediction, and anonymous GPS dispatch.

The Guardian (De Silva D.C.M)

Real-time flight-delay triggers, SLTDA-verified driver matching, and automated SOS cascades.



The Guide (Wanniarachchi P.M.R)

Safety-first A-Star routing, offline AR navigation, and flight-to-hotel bundles.

The Brain (Abinaya R)

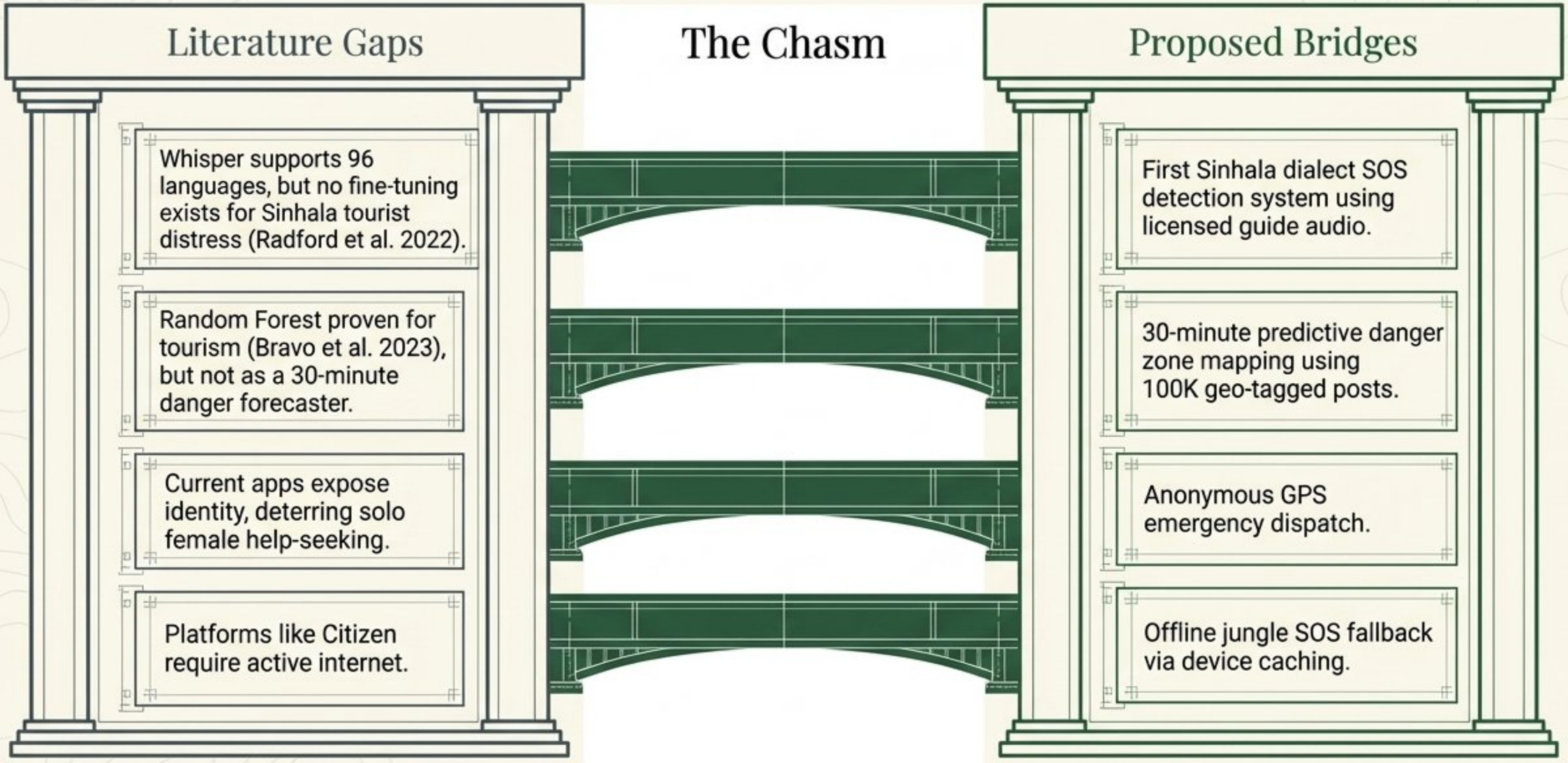
Random Forest crowd prediction, macro-context risk scoring, and green site redirection.

Shared PostgreSQL Database + REST API Contract | SDGs 8, 9, 11



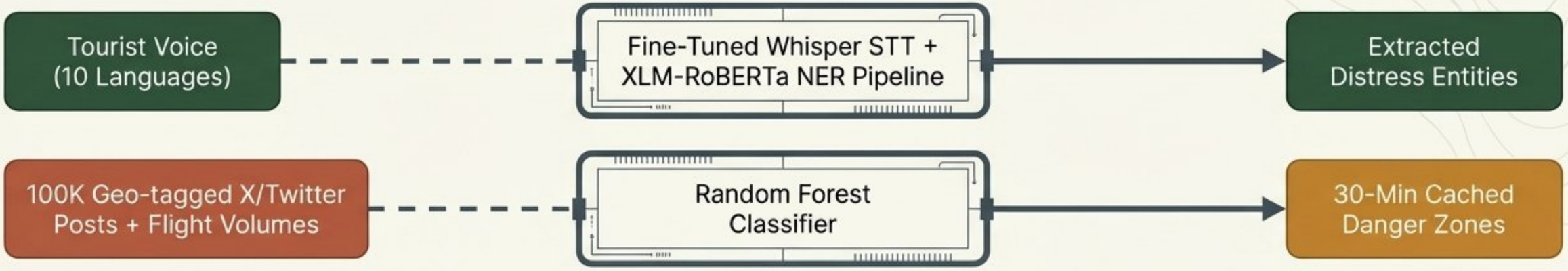
The Ear and Voice: SOS Emergency Detection

Researcher: De Silva D.S.K | IT22108654



Listening for Distress: Architecture and Validation

The NLP Data Pipeline



The Connectivity Fallback Diagram



Evaluation Metrics Matrix

Distress Detection Target	F1-score >= 0.80 (via 5-fold cross-validation)
Danger Zone AUC	Outperforms historical baseline
SOS Response Latency	< 90 seconds in Colombo field trials



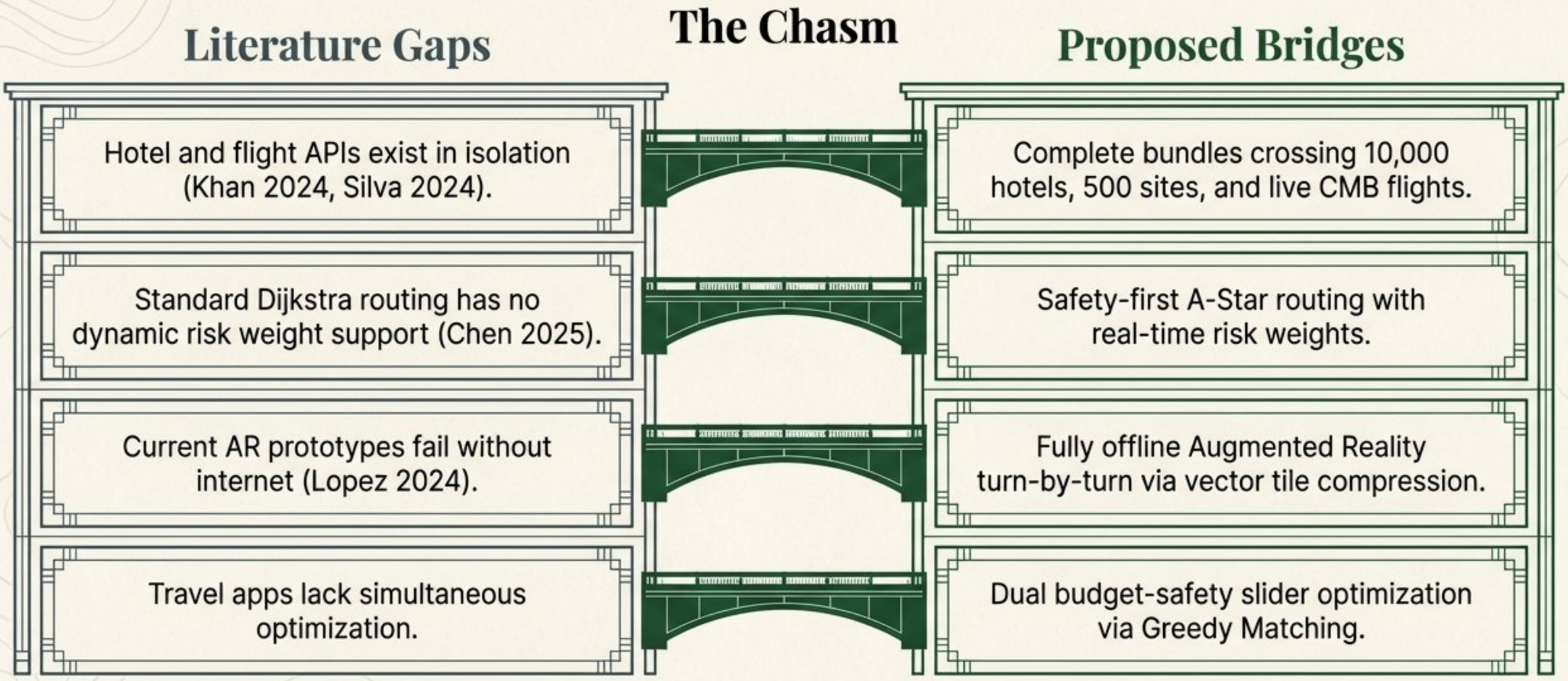
AI Itinerary Generator

Wanniarachchi P.M.R

IT22225092

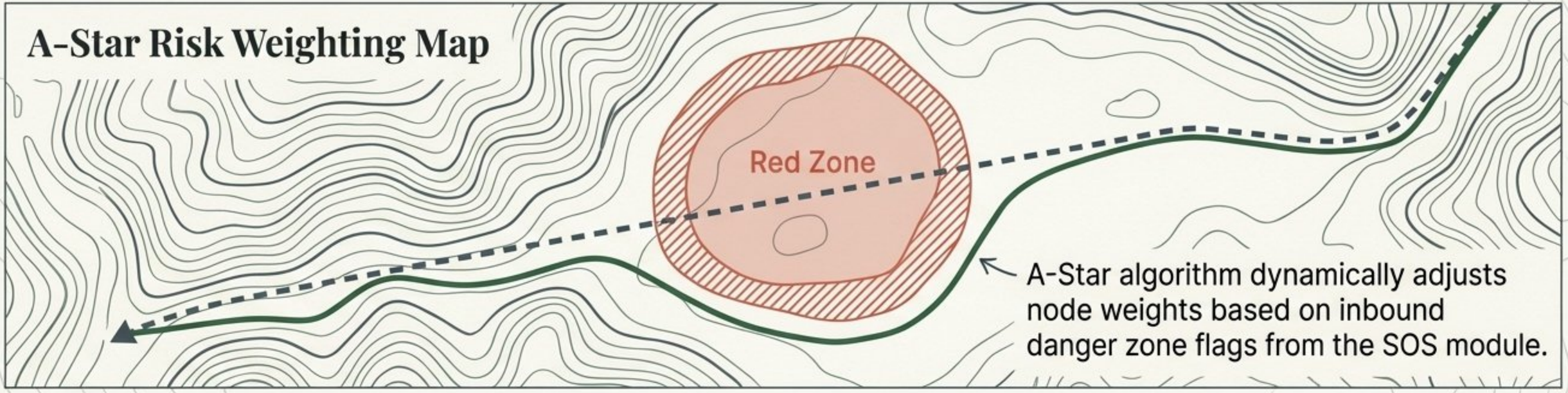
The Guide: AI Itinerary Generator and AR Navigation

Researcher: Wanniarachchi P.M.R | IT22225092

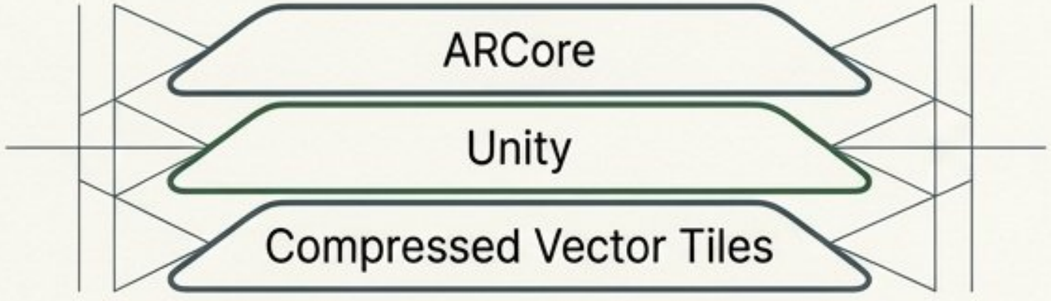


Navigating Safely in the Dark: Risk-Weighted Routing

A-Star Risk Weighting Map



The Offline AR Stack



Capable of handling 100% of complex turn-by-turn navigation in zero-connectivity rural terrain.

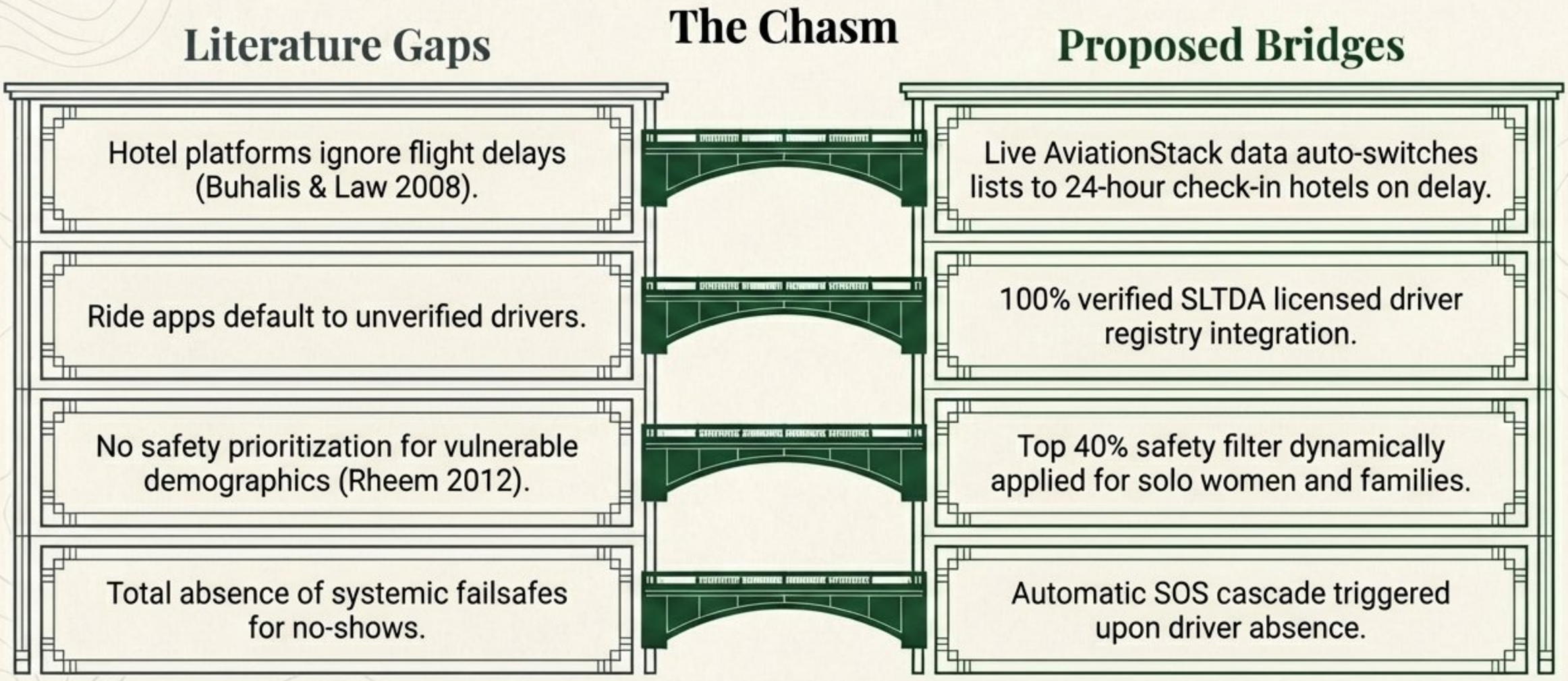
Validation Metrics

- **Danger Avoidance:** 100% simulated avoidance, 30% safer than Google Maps baseline.
- **Planning Speed:** Reduced from >4 hours to <1 hour.
- **AR Accuracy:** 95% correct turn-by-turn in low-connectivity trials.



The Guardian: Hotel-Flight Driver Matcher

Researcher: De Silva D.C.M | IT22108586



Securing the High-Risk Arrival Window

The 45-Minute Cascade



1. Trigger: AviationStack API detects CMB landing.

2. Match: PostgreSQL geospatial geospatial query finds safe hotel < 30 mins away + verifies SLTDA driver.

3. Track: Firebase initiates real-time GPS tracking (30s intervals).



4. Countdown: 45-minute countdown clock begins.

5. Success: Driver GPS pings arrival. Sequence closes.



6. Failure: Clock hits zero with no GPS ping –> RED ALERT –> REST API instantly triggers De Silva D.S.K's SOS system + notifies nearest hotel.

Validation

- **Driver Verification Recall:** 100% strict matching.
- **SOS Trigger Latency:** < 2 minutes from countdown zero to alert dispatch.



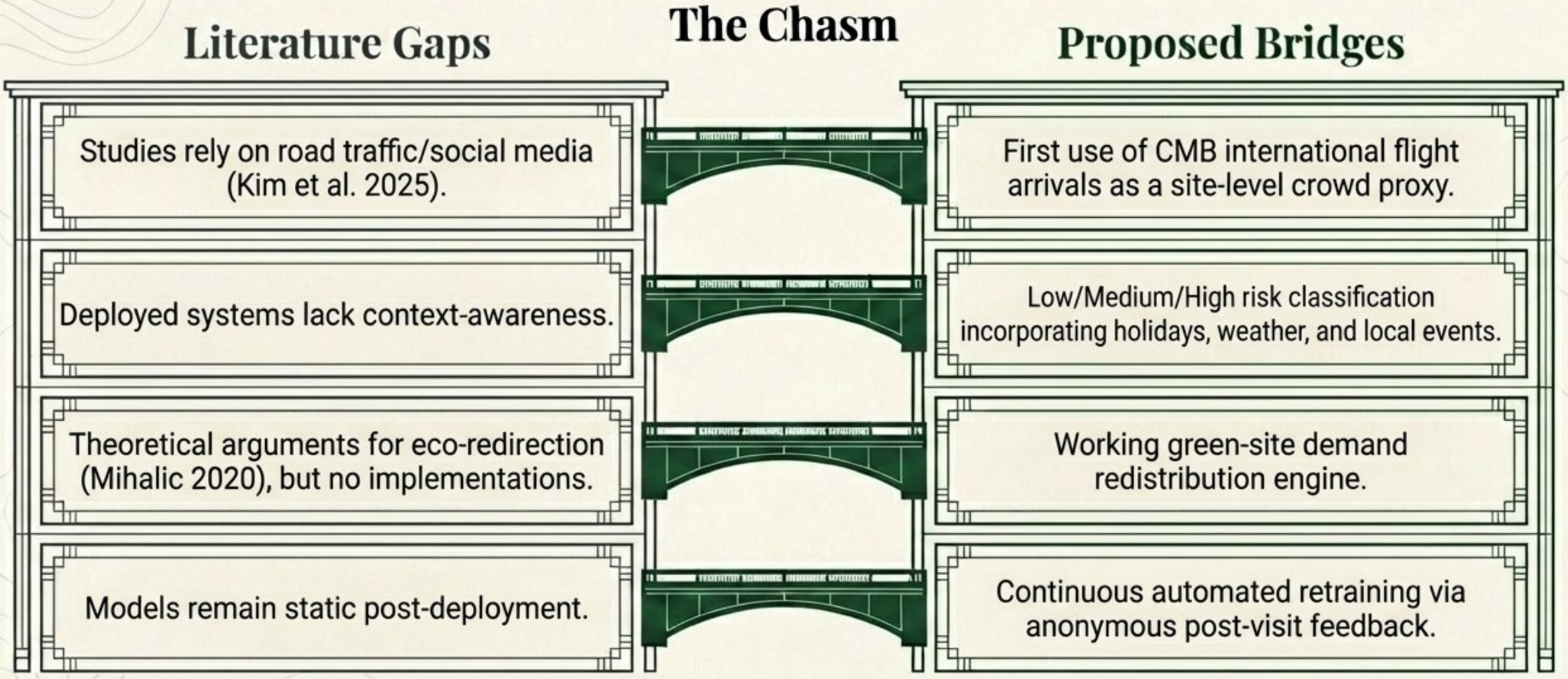
Tourism Risk & Context Intelligence System

Abinaya R

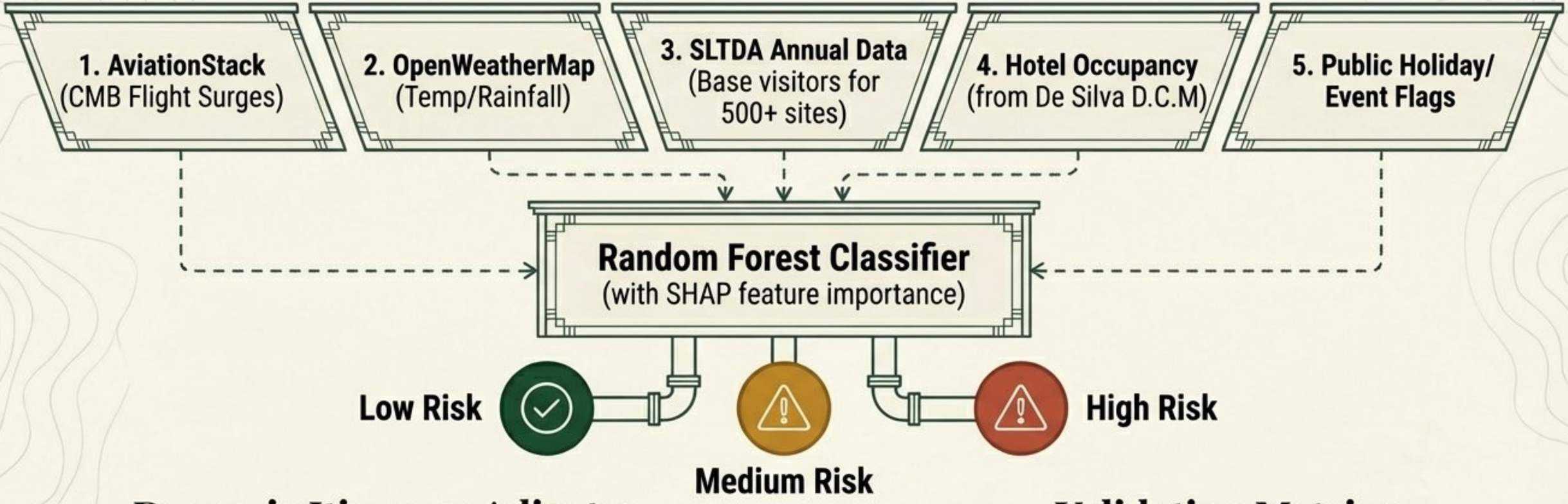
IT22090058

The Brain: Tourism Risk and Context Intelligence

Researcher: Abinaya R | IT22090058



Predicting Systemic Risk: The Context-Aware Funnel



Dynamic Itinerary Adjuster

Sigiriya is High Risk at your planned time.
Switch to [Green Site].

(Triggers API call to Wanniarachchi's routing engine)

Validation Metrics

- Mean Absolute Error (MAE): <= 0.25.
- R-squared (R2): >= 0.70.
- Alert Precision & Recall: >= 0.80.

The Algorithm Showdown: Justifying the Technical Stack

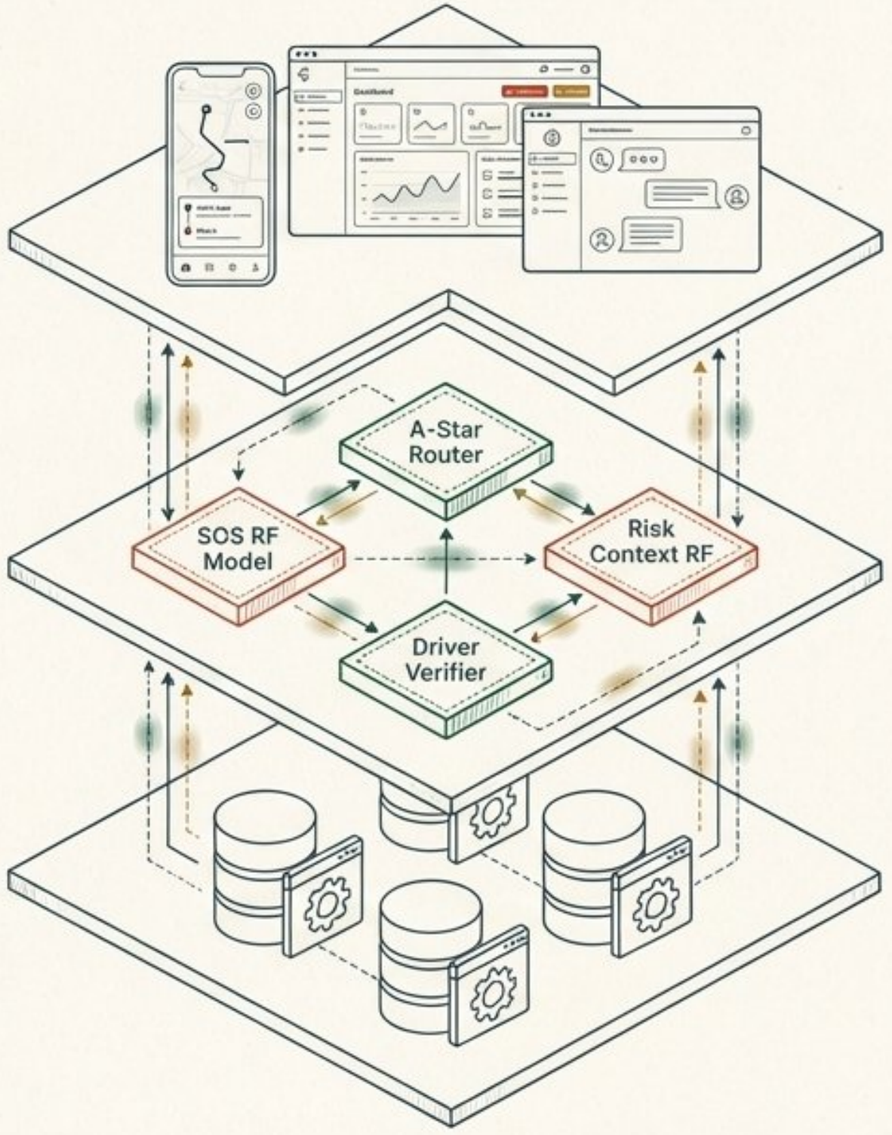
Module Focus	Selected Architecture	Rejected Alternatives	Literature / Technical Justification
Danger Zone & Crowd Risk	Random Forest (De Silva D.S.K, Abinaya R)	CNN / GRU Deep Learning, Linear Regression	Deep learning overfits aggregate SLTDA data (He et al. 2022). RF handles non-linear, mixed-feature data perfectly with high interpretability (SHAP).
Safety Routing	A-Star with Dynamic Risk Weights (Wanniarachchi)	Standard Dijkstra	Dijkstra lacks dynamic risk weight modification necessary for live danger avoidance (Chen 2025).
Voice Detection	Whisper STT (Fine-tuned) + XLM-RoBERTa	Google STT, spaCy	Whisper provides unmatched 96-language baseline; XLM-RoBERTa handles Sinhala token classification natively.
Hotel/Driver Matching	Weighted Geospatial Scoring via PostgreSQL	SVM, Neural Networks	Safety-critical matches require 100% auditable, deterministic rule engines, not probabilistic black boxes.

The Intelligent Topography: Complete Integration Architecture

UI & Dispatch
Flutter App, Operator Dashboard, Police/Hotel Emergency Dispatch.

The Intelligence Engines

The Data Foundation
Unified PostgreSQL Database. Ingesting AviationStack, OpenWeatherMap, SLTDA 500+ Sites, 10,000 Hotels.



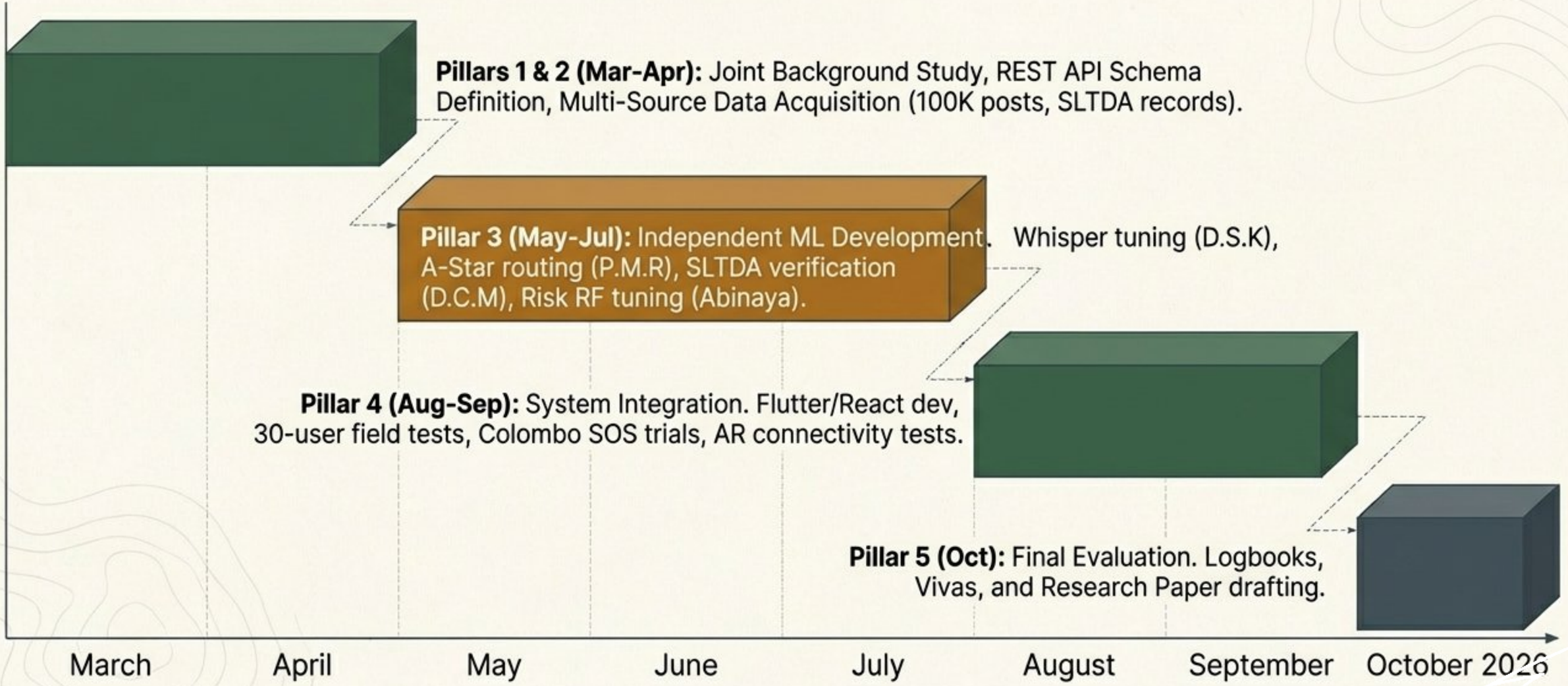
Pulse 1: Abinaya's risk scores alter Wanniarachchi's A-Star routes. →

Pulse 2: Wanniarachchi's itinerary stops trigger De Silva D.C.M's hotel/driver matcher. →

Pulse 3: De Silva D.C.M's no-show triggers De Silva D.S.K's SOS cascade. →

All strictly communicating via predefined REST APIs.

Project Timeline: March to October 2026



Risk Mitigation Framework and Contingencies

Component	Primary Technical Risk	Impact	Mitigation Strategy
SOS Detection	Insufficient Sinhala audio corpus for Whisper fine-tuning.	High	Partner with SLIIT linguistics faculty and Negombo guides; utilize audio augmentation (pitch shift, time stretch, noise addition).
AR Navigation	ARCore GPS drift in rural, uneven terrain.	High	Implement 2D vector map fallback; use major visual anchors (e.g., Sigiriya Rock) for recalibration; 1km safety buffer.
Risk Context	SLTDA provides only annual aggregate data, lacking hourly granularity.	Medium	Engineer simulated intra-day proxy distributions for top 30 sites; strictly label as simulated; validate via field observations.
Platform Integration	API bottlenecks between the 4 modules.	Medium	REST API contracts locked in March; every endpoint developed with mock data to prevent blocking dependencies.

The Academic Bedrock: Core Literature Foundations

SOS & Voice (De Silva D.S.K)

Radford et al. (2023) proves Whisper's 96-language baseline, confirming the viability of fine-tuning for low-resource Sinhala dialects.
He et al. (2022) justifies avoiding Deep Learning for sparse tourist safety datasets.

AR & Routing (Wanniarachchi)

Chen et al. (2025) identifies risk-weighted A-Star routing as an open implementation gap.
Kim & Lee (2023) proves offline vector tile compression is mandatory for mobile navigation.

Matching & Safety (De Silva D.C.M)

Tussyadiah & Pesonen (2016) establishes tourist reliance on authority-verified providers (justifying SLTDA integration).
Rheem (2012) identifies safety as the primary barrier for solo female travelers.

Risk & Context (Abinaya R)

Santos-Rojo et al. (2023) bibliometric study reveals applied technical implementations for overtourism are severely underrepresented.
Bravo et al. (2023) validates Random Forest for heritage site visitor prediction.

Commercialization Strategy: The Market Void

Targeting 3,000,000 arrivals by 2026. One platform replacing five fragmented applications.

Competitor Value Matrix
Google Maps: Provides navigation – ❌ **MISSING:** SOS, verified drivers, proactive risk scoring.
Citizen (US): Provides urban safety – ❌ **MISSING:** Sri Lankan localization, Sinhala support, offline jungle capability.
Booking / Expedia: Provides logistics – ❌ **MISSING:** Crowd intelligence, flight-delay hotel auto-switching.
Sri Lanka Go / Guide-Me: Provides listings – ❌ **MISSING:** Offline AR, dynamic A-Star danger routing.

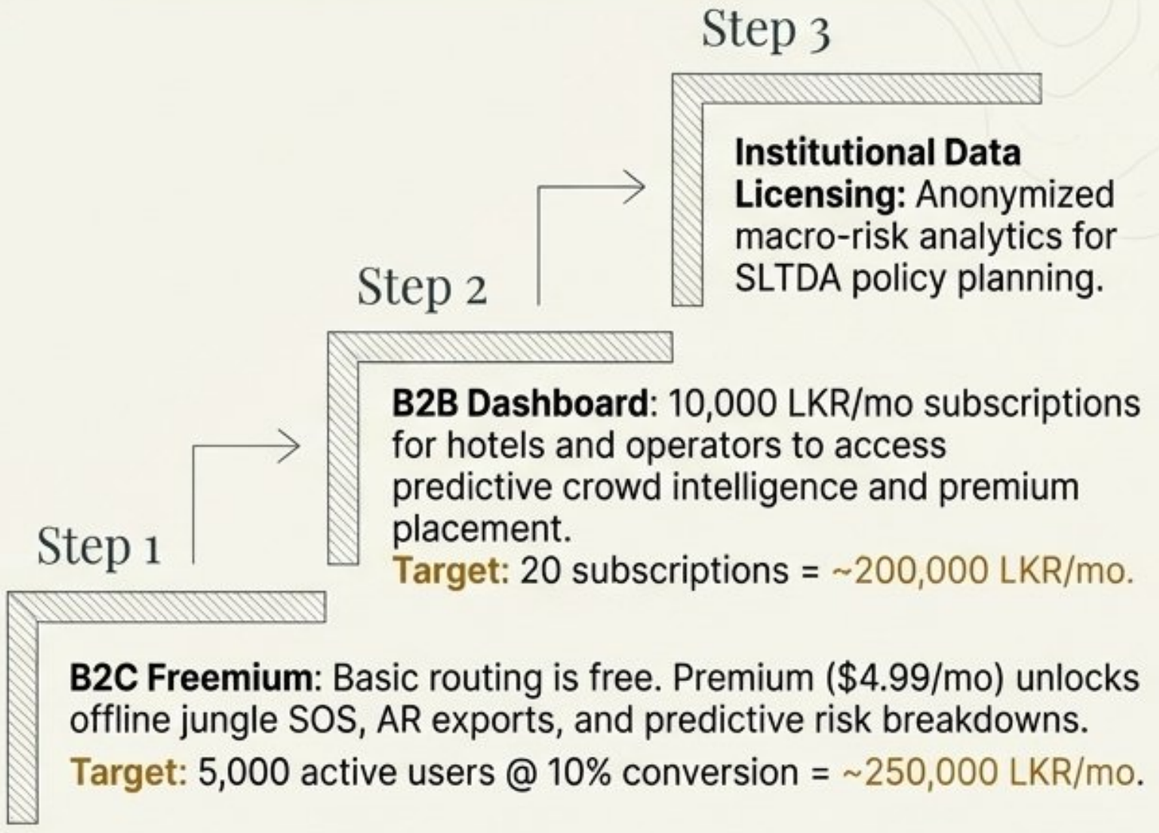
The SafeJourney Advantage: The only platform integrating SLTDA driver registries, localized Sinhala SOS, and predictive 30-minute danger zones in a fully offline-capable mobile architecture.

Financial Viability: Break-Even and Revenue Architecture

Budget & Operating Costs

- Total Development Budget: ~209,300 LKR (Hardware, AviationStack/Cloud API shares, Field Testing).
- Monthly Recurring Operating Costs: ~50,000 LKR (APIs, Maintenance, Cloud Hosting).

Three-Phase Revenue Model



Projected Monthly Revenue: 450,000 LKR. Break-even achieved in Month 1 of commercial operation.

Genuine Alignment: SDGs and AI Research Clusters

Sustainable Development Goals



SDG 8 (Economic Growth): Closing the $1.6B safety-perception revenue gap increases per-day spending and dwell time, directly **expanding local hospitality employment**.

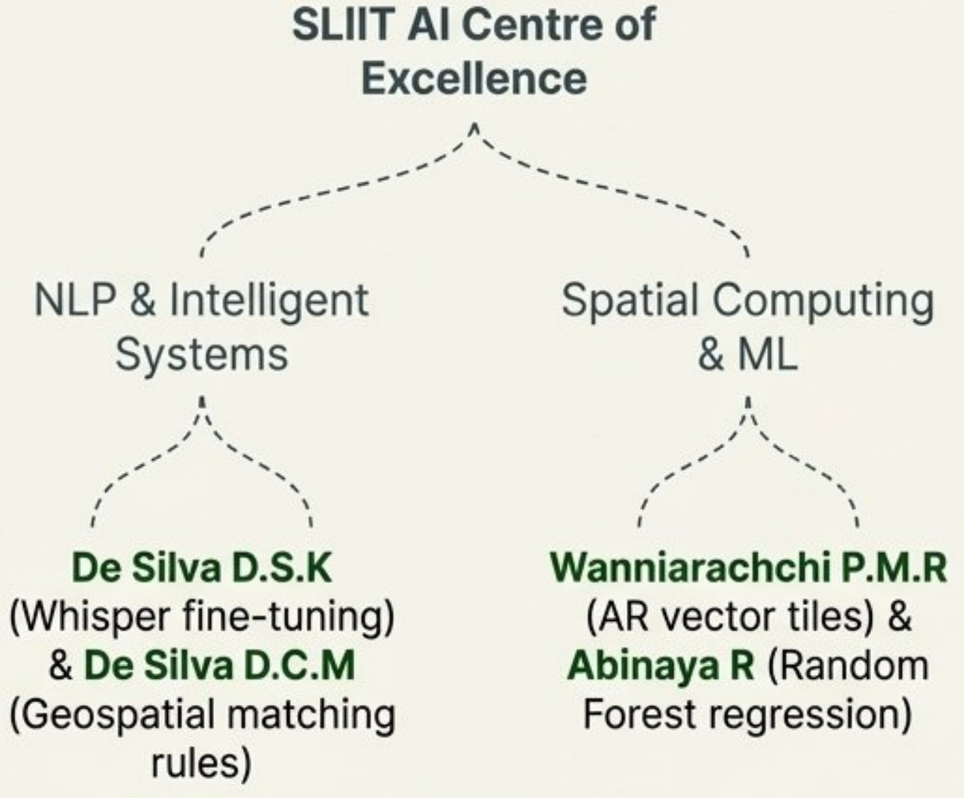


SDG 9 (Innovation): Deploys first Sinhala STT models and offline AR caching, **bridging** Sri Lanka's documented digital tourism **infrastructure deficit**.



SDG 11 (Sustainable Sites): Abinaya R's green site algorithm actively **redirects demand** from degrading UNESCO sites (e.g., Sigiriya, Galle Fort) **to lesser-known, low-risk eco-sites.**

SLIIT AI Centre of Excellence



The Unified Ecosystem: Five Proofs of Impact

1



Quantified Necessity

Directly targets SLTDA's documented **$1.6 Billion revenue gap** and **200+ annual fatalities**.

2



Uncompromising Rigor

Algorithm selections (**Random Forest, A-Star, Whisper**) are comprehensively justified against rejected alternatives (CNN/GRU, Dijkstra) using **40+ peer-reviewed sources**.

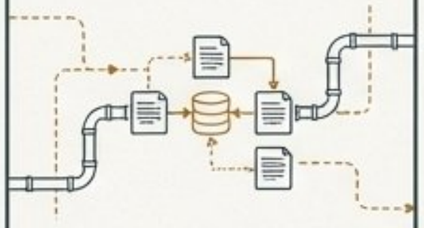
3



Quantifiable Validation

Hard metrics target success: **F1 >= 0.80**, **MAE <= 0.25, R2 >= 0.70**, and **SOS responses under 90 seconds**.

4



True Architecture Integration

Not four siloed papers. A **shared PostgreSQL core** and **strict REST API contracts** ensure every module's output directly improves the others' accuracy.

5



Market Readiness

Clear B2C and B2B monetization pathways ensuring a **Month 1 break-even** on a **~209,300 LKR budget**.